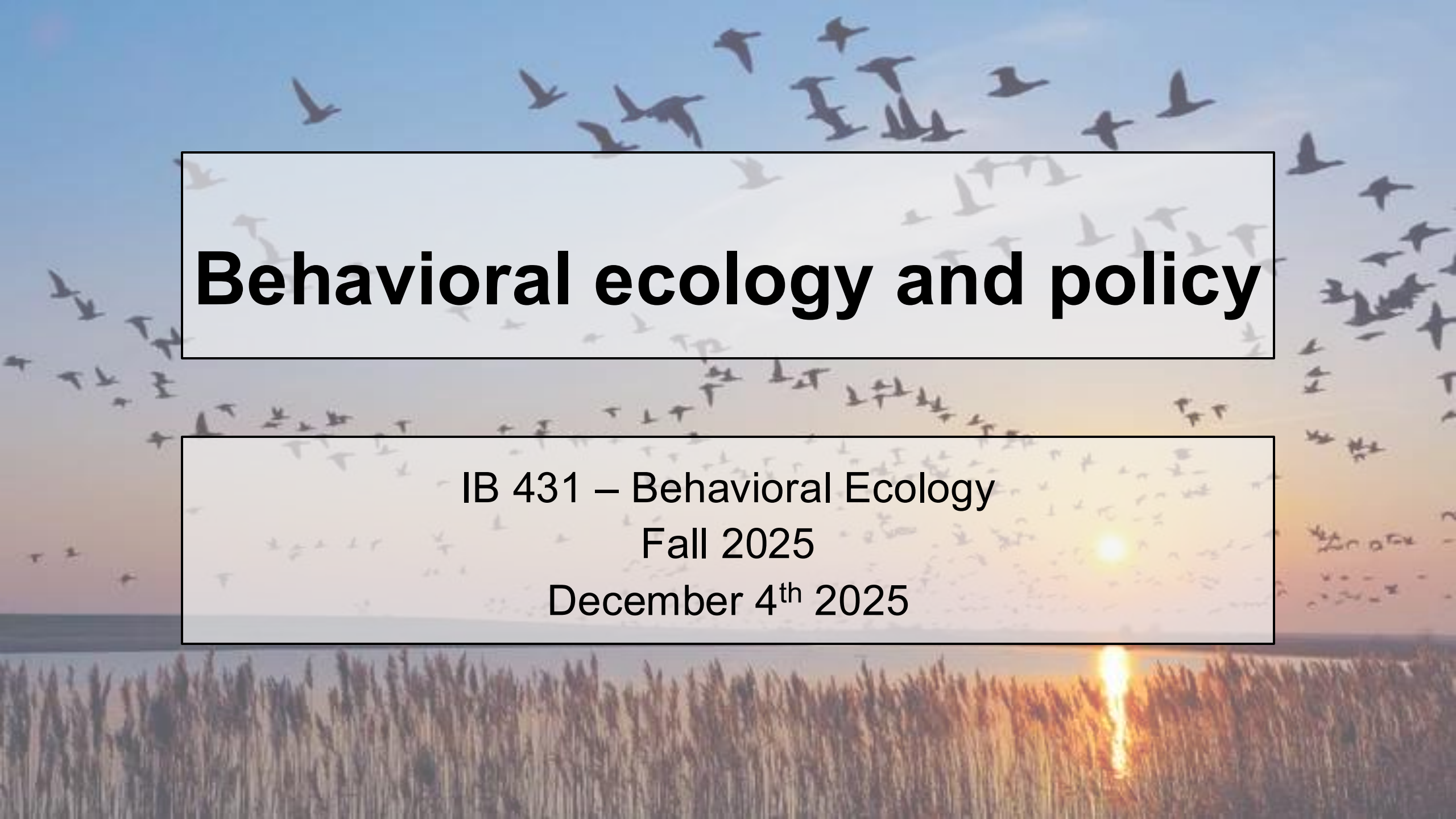


A large flock of birds, possibly geese or swans, is seen in flight against a sky transitioning from blue to orange. Below the birds, a field of tall, golden-brown grass is visible, with a bright sun setting on the horizon, creating a strong reflection on the water in the foreground.

Behavioral ecology and policy

IB 431 – Behavioral Ecology

Fall 2025

December 4th 2025

Announcements

- Last day of class Tuesday – course review, assign final + chance to discuss final questions with classmates and instructors
- Grading of proposals completed – for final version focus on addressing feedback you got
 - Stop by office hours or email us if you need help making figures for proposal and/or responding to feedback

Outline

- I. Course evaluations
- II. Review key court decisions relevant to behavioral ecology
- III. Fur seals game

Behavioral ecology and policy

- Discuss with your tables:
 - In what ways does animal behavior intersect with public policy?
 - Are there any mainstream policy issues that are shaped by animal behavior?
 - Are there any issues that are not mainstream that you think deserve more attention?



Many issues connect with animal behavior

- Conservation and climate change
- Ecosystem services
- Agriculture
- Endangered species
- Fisheries
- Animal welfare
- Public health



House

Senate

Standing Committees

[Agriculture](#)

[Appropriations](#)

[Armed Services](#)

[Budget](#)

[Education and Workforce](#)

[Energy and Commerce](#)

[Ethics](#)

[Financial Services](#)

[Foreign Affairs](#)

[Homeland Security](#)

[House Administration](#)

[Judiciary](#)

[Natural Resources](#)

[Oversight and Government Reform](#)

[Rules](#)

[Science, Space, and Technology](#)

[Small Business](#)

[Transportation and Infrastructure](#)

[Veterans' Affairs](#)

[Ways and Means](#)

[Agriculture, Nutrition, and Forestry](#)

[Appropriations](#)

[Armed Services](#)

[Banking, Housing, and Urban Affairs](#)

[Budget](#)

[Commerce, Science, and Transportation](#)

[Energy and Natural Resources](#)

[Environment and Public Works](#)

[Finance](#)

[Foreign Relations](#)

[Health, Education, Labor, and Pensions](#)

[Homeland Security and Governmental Affairs](#)

[Judiciary](#)

[Rules and Administration](#)

[Small Business and Entrepreneurship](#)

[Veterans' Affairs](#)

Animal behavior may have unexpected relevance

- Economics
 - Agricultural pests like locusts can crash local economies
- Military
 - US Army Corps of Engineering oversees wetland protection
- Transportation
 - Bird migration and plane schedules
 - Wildlife crossings



Important legal cases for behavioral ecology

- Conservation and climate change
- Ecosystem services
- Agriculture
- Endangered species
- Fisheries
- Animal welfare
- Public health



Clean Water Act

- Passed in 1972 (366-11; 74-0)
- Establishes the basic structure for regulating discharges of pollutants into the **waters of the United States**



What are “waters of the United States”?

- “Navigable waters”
(Section 404, CWA)
 - Including many intermittent lakes, streams, swamps, other wetlands



What are “waters of the United States”?

- *United States vs Riverside Bayview Homes* (1985)
 - Developer tried filling wetlands, state court and ultimately Supreme Court ruled against them



Migratory bird rule

- Typically, isolated, intrastate, non-navigable waters were not protected under the CWA
- Migratory bird rule (1986) extends protection if waters are used by protected species
- Birds cross state lines = makes it a federal concern (Commerce Clause of the Constitution)



Solid Waste Agency of Northern Cook County v. Army Corps of Engineers (2001)

- Cook County wanted to build a landfill over an isolated wetland
- Court knocked down migratory bird rule (5-4 ruling)
- Animal behavior played a key role in arguments that would shape numerous aspects of the economy, transportation, agriculture, and urban development

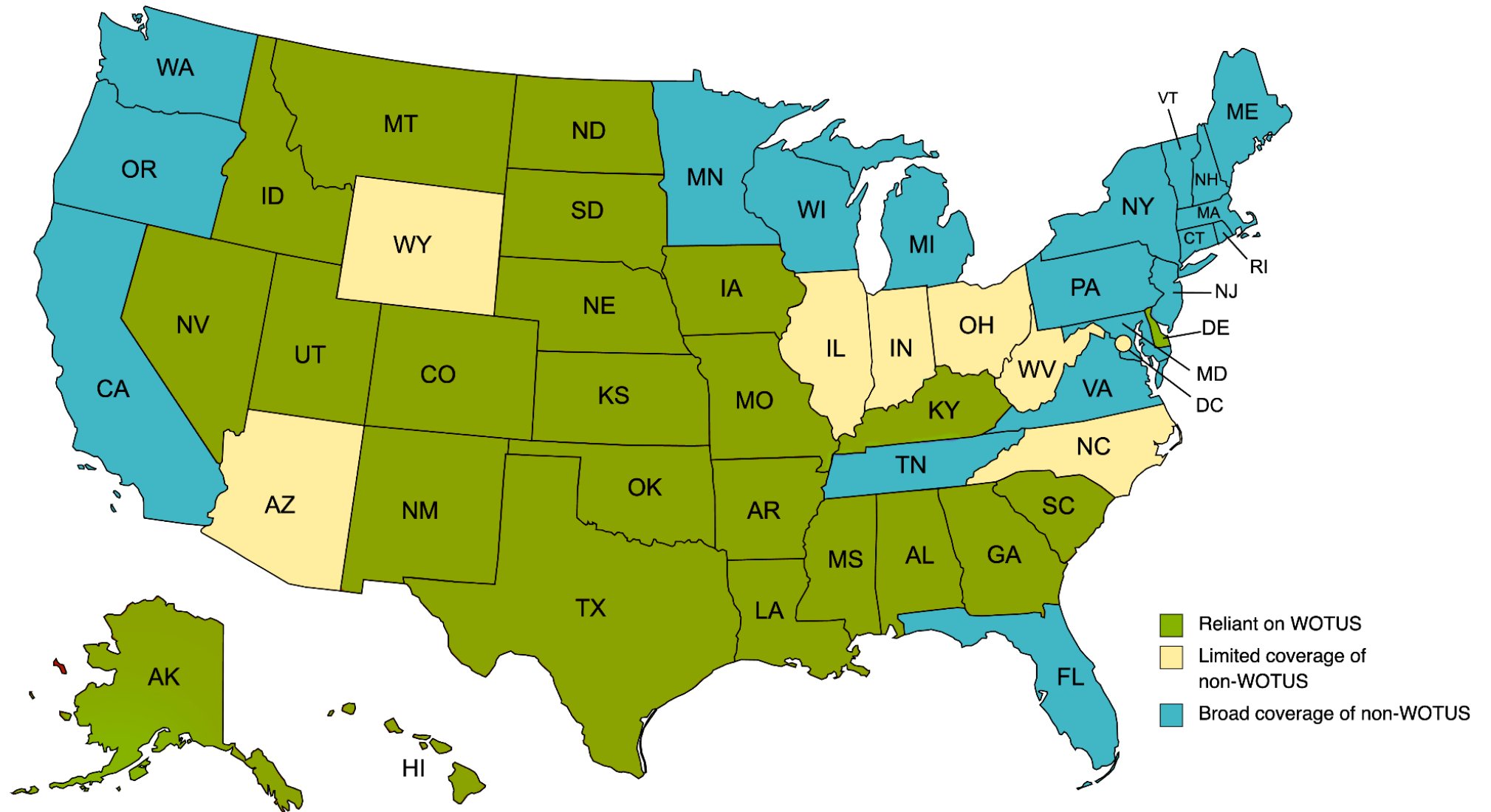


What are “waters of the United States”?

- *Rapanos vs United States* (2006)
 - Challenged the 1985 ruling, generally reduced protection
 - Justice Scalia (plurality) – CWA only allows federal jurisdiction over “relatively permanent, standing or **continuously flowing bodies of water** 'forming geographic features'” or having a continuous surface connection to navigable waters
 - Justice Kennedy (concurring) – also should include wetlands with a "**significant nexus**" to navigable waters

Sackett vs. EPA (2023)

- Supreme Court rejects the “significant nexus” concept
- Decides CWA covers only wetlands and permanent bodies of water with a "continuous surface connection" to "traditional interstate navigable waters”
- Significantly limited wetland protection – up to 60 million acres may lose CWA protection and will rely on state action



24 states were reliant on CWA for wetland protection

Endangered Species Act

- Signed into law by Richard Nixon in 1973, three years after he formed the EPA
- Still one of the most powerful laws that regulate environmental protection



Endangered Species Act

- Court's first interpretation:
Tennessee Valley Authority v. Hill (1978)
 - Ruled that dam construction would damage critical habitat for snail darters (6-3 vote)



Snail darter (*Percina tanasi*)

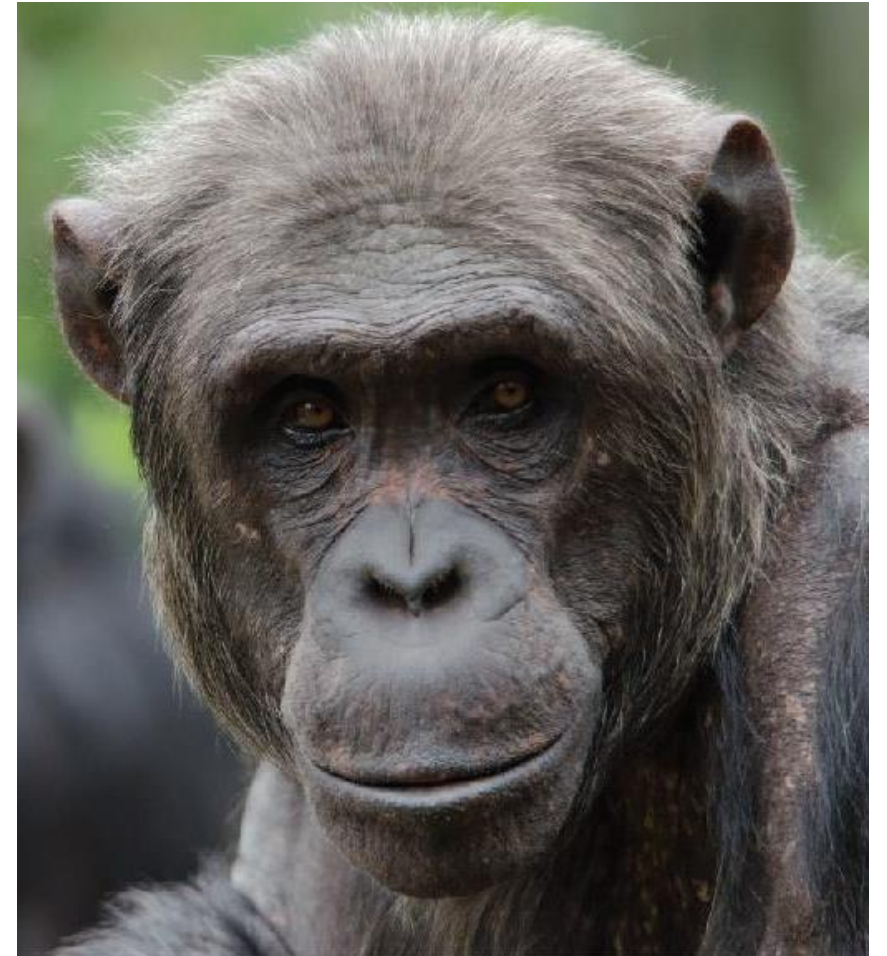
Babbitt v. Sweet Home Chapter, Communities for a Great Oregon (1995)

- ESA requires that no person "take" an endangered or threatened species (take = harm, kill, harass)
- Court determines that habitat destruction counted as "taking" an endangered species (6-3 decision)



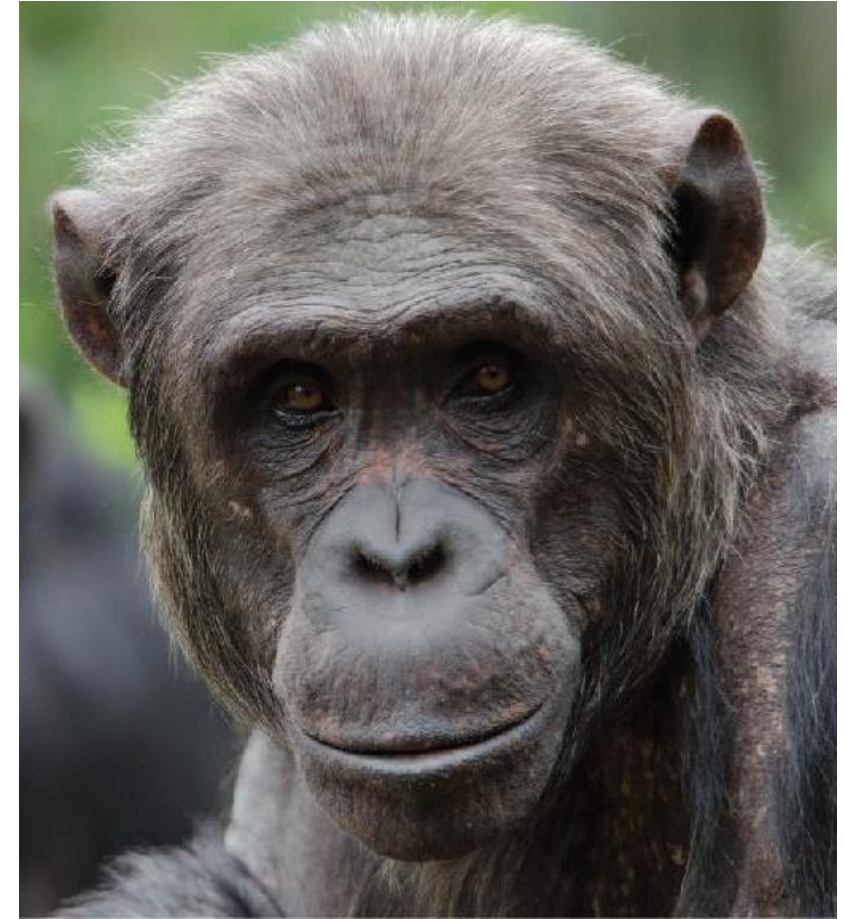
Habeas corpus for chimpanzees?

- *Nonhuman Rights Project v. Lavery* (New York State Supreme Court; 2018)
 - Filed petitions for two chimpanzees (Tommy and Kiko) who had been confined to small cages by “owners”
 - Argued that they should have legal personhood and be able to have right to protest wrongful detainment



“Unlike human beings, chimpanzees cannot bear any legal duties, submit to societal responsibilities or be held legally accountable for their actions. In our view, it is this incapability to bear any legal responsibilities...that renders it inappropriate to confer upon chimpanzees legal rights...such as the right to liberty protected by the writ of habeas corpus — that have been afforded to human beings”

“The ability to acknowledge a legal duty or legal responsibility should not be determinative of entitlement to habeas relief, since, for example, infants cannot comprehend that they owe duties or responsibilities and a comatose person lacks sentience, yet both have legal rights.”



Discuss with your tables. Do you think chimpanzees deserve legal rights like habeas corpus?



MARCH 23, 2022 | ANIMAL LAW & POLICY PROGRAM, MEDIA RELEASE

Constitutional Court of Ecuador Recognizes Animal Rights in Landmark Ruling

For immediate release: 23 March, 2022

Appeals court rejects habeas corpus for chimpanzees, lawyers will appeal to Michigan Supreme Court

BY: KATHERINE DAILEY - OCTOBER 20, 2025 4:00 PM



From L-R: Michigan Court of Appeals Judges Matthew Ackerman, Brock Swartzle and Christopher Trebilcock. Oct. 14, 2025 | Screenshot

Monkey selfie copyright dispute

- Wildlife photographer had macaque steal his camera and take selfies
- Copyright dispute – Wikimedia Commons said they could use it because photographer did not take it and the monkey could not hold a copyright
- PETA argues it should belong to the macaque
 - Appeals court rules against PETA



Diamond v. Chakrabarty (1980)

- Geneticist who had developed bacteria that would treat oil spills tried to file patent but living things were not covered
- Supreme Court decided in favor, allowed for living things to be patented



Diamond v. Chakrabarty (1980)

- Geneticist who had developed bacteria that would treat oil spills tried to file patent but living things were not covered
- Supreme Court decided in favor, allowed for living things to be patented
- **Should living things be able to be patented?**



Winter v. Natural Resources Defense Council (2008)

- Supreme Court rules that navy is allowed to use sonar freely despite possible effects on marine mammals

"the most serious possible injury would be harm to an unknown number of marine mammals that they study and observe...forcing the Navy to deploy an inadequately trained antisubmarine force jeopardizes the safety of the fleet"



Many issues connect with animal behavior

- Conservation and climate change
- Ecosystem services
- Agriculture
- Endangered species
- Fisheries
- Animal welfare
- Public health



Outline

- I. Course evaluations
- II. Review key court decisions relevant to behavioral ecology
- III. Fur seals game**



Fur Seal Game

Male fur seals engage in violent battles over control of beaches and access to females.

Each table will have a collection of chips. These represent available mates on the island.

You must choose which island you will go to and which strategy you will be adopting.



Fur Seal Game

You may either fight or sneak.

If you fight, you and all other fighters will roll a die.

If you sneak, you will take a single chip from the table.

If you are a fighter and roll the highest number, you get all the remaining chips.



In the Wild

Reproductive systems with uneven parental care usually lead to this kind of reproductive skew.

Most members of the population get few or none reproductive events, while a handful of winners receive far more proportionally.

Joseph I. Hoffman, Ian L. Boyd, William Amos, MALE REPRODUCTIVE STRATEGY AND THE IMPORTANCE OF MATERNAL STATUS IN THE ANTARCTIC FUR SEAL *ARCTOCEPHALUS GAZELLA*, *Evolution*, Volume 57, Issue 8, 1 August 2003, Pages 1917–1930, [https://doi-org.proxy2.library.illinois.edu/10.1111/j.0014-3820.2003.tb00598.x](https://doi.org.proxy2.library.illinois.edu/10.1111/j.0014-3820.2003.tb00598.x)

